

Department of Mechanical and Production Engineering

Aarhus University

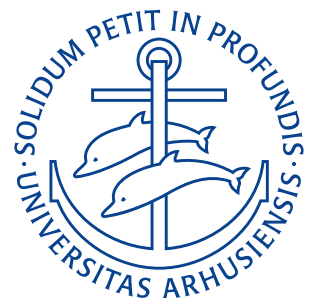
Take-Home Assignment Lec. 18–21

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Course: Statics & Strength of Materials — 290221U006

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Problem 16.17

Determine the maximum deflection of the beam. EI is constant.

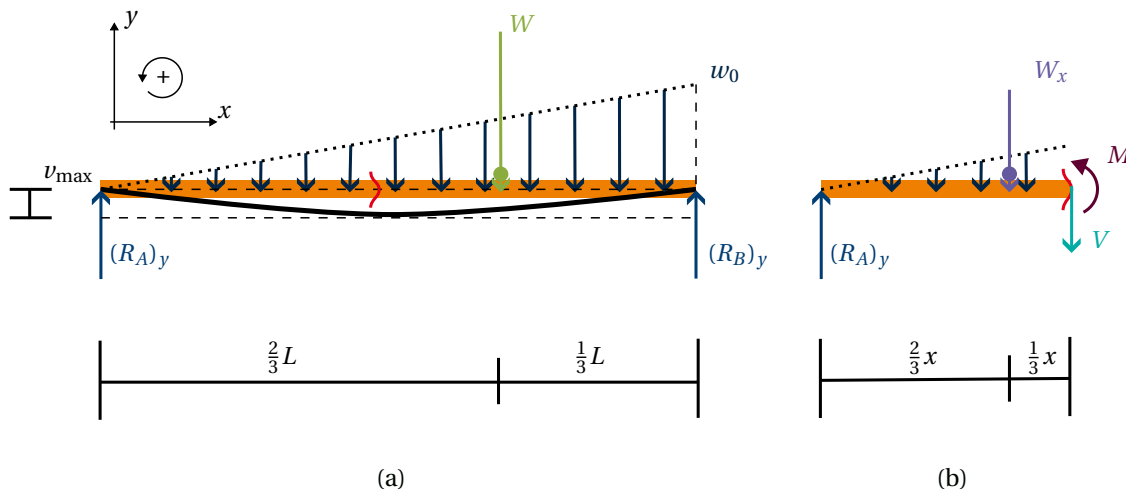
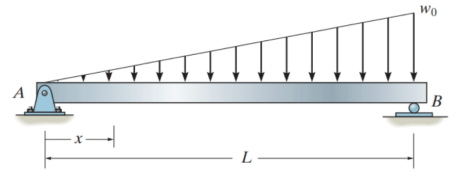


Figure 1.1: (a) Free body diagram of the beam and (b) a cut of said beam taken at a distance x .

Derivation

Assuming the load varies linearly, which looks to be the case based on the figure, the weight w as a function of distance x is

$$w(x) = \frac{x}{L} w_0.$$

If we take a cut off the figure at distance x , (as shown on Figure 1.1b) the resultant on the left hand side is

$$W_x = \frac{1}{2} \frac{x}{L} w_0 \cdot x = \frac{x^2}{2L} w_0.$$

This will act at $1/3x$ from the cut towards A. A moment equilibrium about the cut gives

$$\begin{aligned} 0 &= \sum M_x \\ 0 &= -(R_A)_y x + W_x \frac{x}{3} + M \\ M(x) &= (R_A)_y x - W_x \frac{x}{3} \\ &= (R_A)_y x - \frac{x^3}{6L} w_0. \end{aligned}$$

Now we will take a moment equilibrium about point B to obtain the reaction force $(R_A)_y$. The total distributed load on the beam is

$$W = \frac{1}{2} w_0 L$$

which acts at $1/3L$ from B . Hence,

$$\begin{aligned} 0 &= \sum M_B \\ 0 &= -(R_A)_y L + W \frac{L}{3} \\ (R_A)_y &= \frac{W}{3} \\ &= \frac{L}{6} w_0. \end{aligned}$$

Inserting this into the expression for the internal moment as a function of distance $M(x)$ from before, we obtain

$$M(x) = \frac{w_0 L}{6} \cdot x - \frac{w_0}{6L} x^3.$$

For a beam of constant flexural rigidity (EI) a common result is

$$EI \frac{d^2 v}{dx^2} = M(x).$$

By insertion and subsequent integration, we obtain:

$$\begin{aligned} EI \frac{d^2 v}{dx^2} &= M(x) = \frac{w_0 L}{6} x - \frac{w_0}{6L} x^3 \\ EI \frac{dv}{dx} &= \int M(x) dx = \frac{w_0 L}{12} x^2 - \frac{w_0}{24L} x^4 + C_1 \\ EI \cdot v(x) &= \int \int M(x) dx dx = \frac{w_0 L}{36} x^3 - \frac{w_0}{120L} x^5 + C_1 x + C_2. \end{aligned}$$

As the ends are fixed, our boundary conditions are

$$v(0) = 0, \quad v(L) = 0.$$

For $v(0) = 0$ to hold, C_2 must be zero. At $x = L$, we then get:

$$\begin{aligned} 0 &= \frac{w_0 L}{36} L^3 - \frac{w_0}{120L} L^5 + C_1 L \\ \Rightarrow C_1 &= -\frac{7}{360} w_0 L^3. \end{aligned}$$

Hence,

$$EI \cdot v(x) = \frac{w_0 L}{36} x^3 - \frac{w_0}{120L} x^5 - \frac{7 w_0 L^3}{360} x.$$

To find the point of largest deflection we must find the extremum of this function. Hence, we set the derivative equal to zero as (note EI is ignored here, as from the above equation it can be seen to solely scale the deformation $v(x)$ and will therefore not matter when determining the point of maximum deformation):

$$\begin{aligned} 0 &= \frac{dv(x)}{dx} \\ 0 &= \frac{w_0 L}{12} x^2 - \frac{w_0}{24L} x^4 - \frac{7 w_0 L^3}{360}. \end{aligned}$$

Solving the above 4th degree polynomial gives the following 4 solutions (assuming $L \neq 0$ and $w_0 \neq 0$):

$$\begin{aligned} x_1 &\approx -1,32L \\ x_2 &\approx -0,519L \\ x_3 &\approx 0,519L \\ x_4 &\approx 1,32L. \end{aligned}$$

Of these, only the solution $x_3 \approx 0,519L$ can be the correct solution, as the other three values all lie at a point not

on the beam. Substituting this solution of $x = x_3$ into our expression for $v(x)$ we obtain:

$$v_{\max} = v(x_3) \approx -6,52 \cdot 10^{-3} \frac{L^4 w_0}{EI}.$$

$$v_{\max} \approx -6,52 \cdot 10^{-3} \frac{L^4 w_0}{EI}$$

RESULT

Problem 16.18

Determine the maximum slope of the beam. EI is constant.

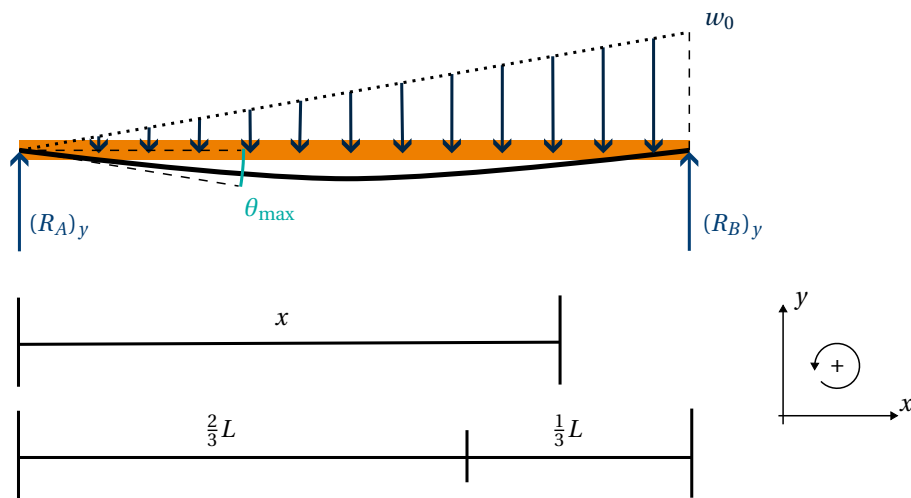
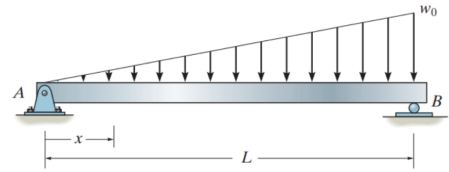


Figure 2.1: Free body diagram of the beam.

Derivation

From Problem 16.17 we have the results:

$$EI \frac{d^2 v(x)}{dx^2} = \frac{w_0 L}{6} x - \frac{w_0}{6L} x^3$$

$$EI \frac{dv(x)}{dx} = \frac{w_0 L}{12} x^2 - \frac{w_0}{24L} x^4 - \frac{7w_0 L^3}{360}.$$

As $d^2 v/dx^2 = d\theta/dx$ and $dv/dx = \theta$ the above can also be written as:

$$EI \frac{d\theta(x)}{dx} = \frac{w_0 L}{6} x - \frac{w_0}{6L} x^3$$

$$EI \theta(x) = \frac{w_0 L}{12} x^2 - \frac{w_0}{24L} x^4 - \frac{7w_0 L^3}{360}.$$

Finding the maximum slope $\theta(x)$ corresponds to finding the value of x for which $d\theta(x)/dx = 0$. Hence, assuming, $L \neq 0$

$$0 = \frac{w_0 L}{6} x - \frac{w_0}{6L} x^3$$

$$\Rightarrow x_1 = 0, \quad x_2 = L.$$

Solving the equation for the slope for these values, we obtain:

$$\theta(x) = \frac{1}{EI} \left(\frac{w_0 L}{12} x^2 - \frac{w_0}{24L} x^4 - \frac{7w_0 L^3}{360} \right)$$

$$\theta(x_1) = \frac{1}{EI} \left(-\frac{7w_0L^3}{360} \right) = -\frac{7w_0L^3}{360EI}$$
$$\theta(x_2) = \frac{w_0L^3}{EI} \left(\frac{1}{12} - \frac{1}{24} - \frac{7}{360} \right) = \frac{w_0L^3}{45EI}.$$

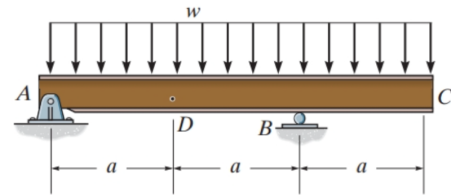
As $|-7/360| < |1/45|$ the maximum deflection occurs at $x_2 = L$, where:

$$\theta_{\max} = \frac{w_0L^3}{45EI}$$

RESULT

Problem 16.36

Determine the slope at A and the deflection at point D of the overhang beam. EI is constant.



Derivation

As the load varies linearly with displacement (constant distributed load) and the load is not assumed large enough to significantly distort the original geometry of the beam, the method of superposition can be employed. Using this, we split up the beam into the two sections shown on Figure 3.1 and Figure 3.2. One of these is the simply supported beam AB with the uniform load w and the other being the effect on AB from the uniform load on the overhang BC .

Simply supported beam AB

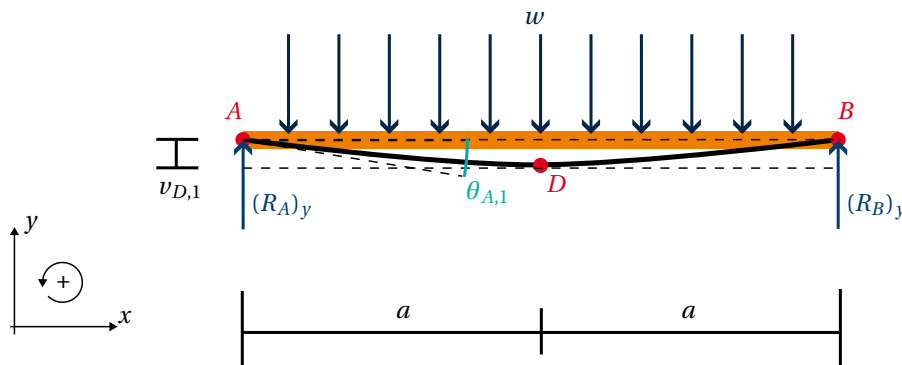


Figure 3.1: Free Body diagram of the simply supported beam AB .

For the simply supported beam, depicted on Figure 3.1, Appendix D gives the slope at A as:

$$\theta_{A,1} = -\frac{wL^3}{24EI} = -\frac{w(2a)^3}{24EI} = -\frac{wa^3}{3EI}.$$

The deflection at point D will, due to symmetry, be the maximum deflection. From Appendix D, this is seen to be

$$v_{D,1} = -\frac{5wL^4}{384EI} = -\frac{5w(2a)^4}{384EI} = -\frac{5wa^4}{24EI}.$$

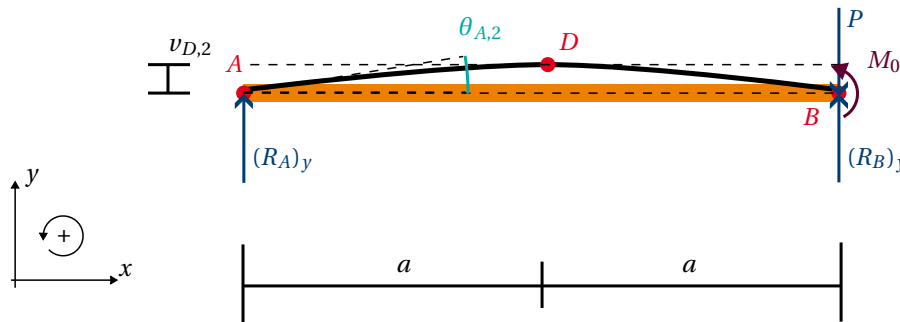


Figure 3.2: Free Body diagram of the contribution from the overhang BC.

Contribution from overhang BC

We will replace the distributed load over BC with a corresponding couple moment and a force acting downwards on R_B as shown on Figure 3.2. The equivalent moment M_0 is then given by

$$M_0 = -w \cdot a \cdot \frac{1}{2} \cdot a = -\frac{1}{2} w a^2.$$

From the Appendix D the slope at A caused by the moment from the overhang BC is

$$\theta_{A,2} = \frac{-M_0 L}{6EI} = -\frac{(-\frac{1}{2} w a^2) 2a}{6EI} = \frac{w a^3}{6EI}.$$

And the deflection is found from the elastic curve given in the Appendix as

$$v(x) = \frac{-M_0 x}{6EIL} (L^2 - x^2) = \frac{w a x}{24EI} (4a^2 - x^2).$$

Evaluating this at $x = a$ gives the deflection for D as

$$v(a) = v_{D,2} = \frac{w a^2}{24EI} (4a^2 - a^2) = \frac{w a^4}{8EI}.$$

Principle of superposition

Using the principle of superposition we find the total slope at A as the algebraic sum of the slopes caused by either section, as:

$$\theta_A = \theta_{A,1} + \theta_{A,2} = -\frac{w a^3}{3EI} + \frac{w a^3}{6EI} = -\frac{w a^3}{6EI}.$$

Likewise, the total deflection at D will be the algebraic sum of the deflections at D caused by either of the two sections:

$$v_D = v_{D,1} + v_{D,2} = -\frac{5w a^4}{24EI} + \frac{w a^4}{8EI} = -\frac{w a^4}{12EI}.$$

$$\begin{aligned}\theta_A &= -\frac{wa^3}{6EI} \\ v_D &= -\frac{wa^4}{12EI}.\end{aligned}$$

RESULT

Problem 16.46

The assembly consists of a cantilevered beam CB and a simply supported beam AB . If each beam is made of A-36 steel and has a moment of inertia about its principal axis of $I_x = 50 \cdot 10^6 \text{ mm}^4$, determine the displacement at the center D of beam BA .

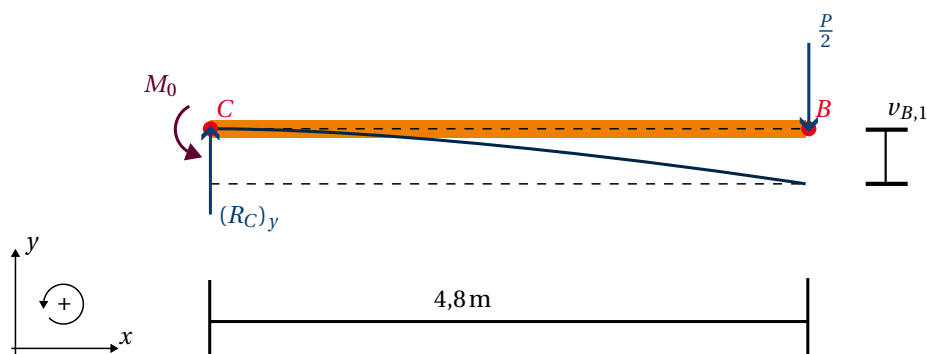
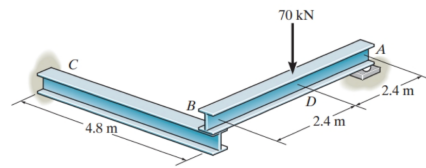


Figure 4.1: Free Body diagram of the cantilevered beam BC .

Derivation

Due to symmetry, it must be true that $R_A = R_B = R_C = -P/2 = 35 \text{ kN}$. As was the case in [Problem 16.36](#) we will once again use the principle of superposition. This time we split the system up into subdivisions consisting of member BC and AB respectively.

Member BC is a cantilevered beam, shown on [Figure 4.1](#). From Appendix D, the maximum displacement of this (which will be at point B) is

$$v_{B,1} = -\frac{\frac{P}{2}L^3}{3EI} = -\frac{PL^3}{6EI}.$$

If we do not consider the deflection caused by the load the member AB will simply rotate rigidly between A and B as depicted on [Figure 4.2](#). I.e. if B “sinks” by a distance of $-PL^3/(6EI)$, then the midpoint D of AB will “sink” half this distance:

$$v_{D,1} = \frac{1}{2} \left(-\frac{PL^3}{6EI} \right) = -\frac{PL^3}{12EI}.$$

For member AB , depicted on [Figure 4.3](#), Appendix D gives a deflection of

$$v_{D,\text{load}} = -\frac{PL^3}{48EI}.$$

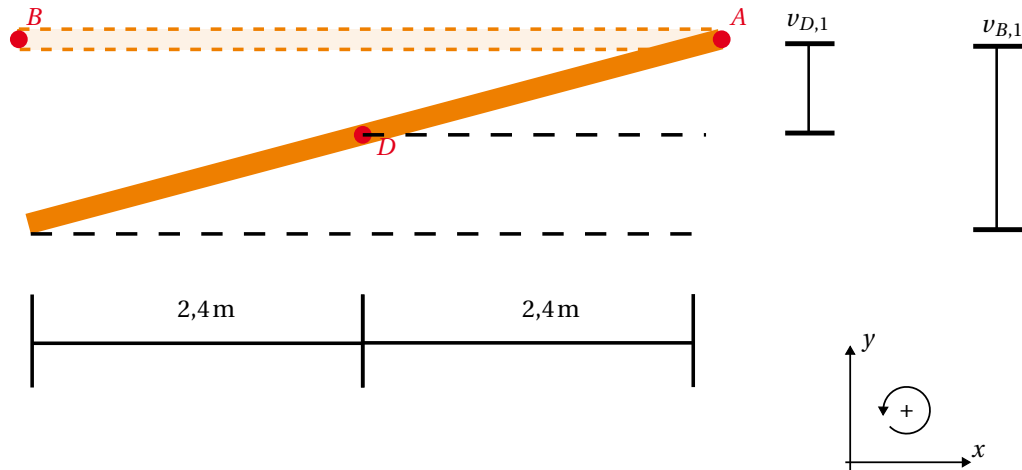


Figure 4.2: Free Body diagram of the rigid rotation of member AB.

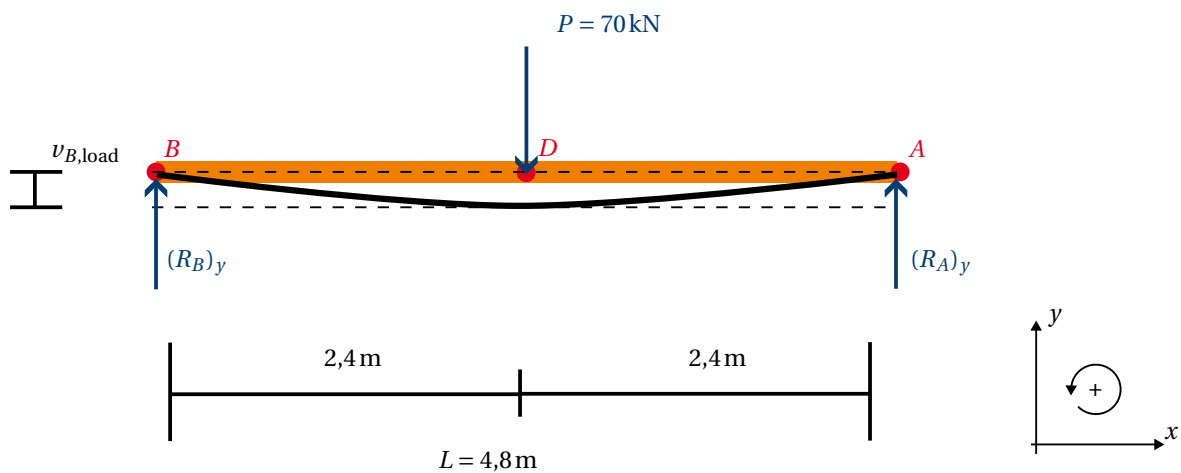


Figure 4.3: Free body diagram of the deflection of member AB.

From the principle of superposition, the total deflection at D will therefore be:

$$v_D = v_{D,1} + v_{D,\text{load}} = -\frac{PL^3}{12EI} - \frac{PL^3}{48EI} = -\frac{5PL^3}{48EI}.$$

From the figure we see that $P = 70 \text{ kN}$ and $L = 4,8 \text{ m}$. Furthermore $I_x = 50 \cdot 10^6 \text{ mm}^4$ is given. In the course literature the elastic modulus of A-36 steel is given as $E = 200 \text{ GPa}$. Hence

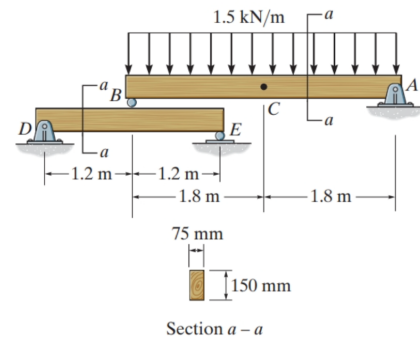
$$v_D = -\frac{5 \cdot 70 \text{ kN} \cdot (4,8 \text{ m})^3}{48 \cdot 200 \text{ GPa} \cdot 50 \cdot 10^6 \text{ mm}^4} = -8,06 \text{ cm}.$$

$$v_D = -8,06 \text{ cm}$$

RESULT

Problem R16.7

Using the method of superposition, determine the displacement at C of beam AB . The beams are made of wood having a modulus of elasticity of $E = 10 \text{ GPa}$.



Derivation

Here, we will utilize the same idea of using the principle of superposition as was the case in [Problem 16.46](#).

Due to symmetry we have that

$$R_B = R_A = -\frac{1}{2}W.$$

The equivalent load W that corresponds to our distributed load $w = 1,5 \frac{\text{kN}}{\text{m}}$ is

$$W = w \cdot L_{AB} = -1,5 \frac{\text{kN}}{\text{m}} \cdot 3,6 \text{ m} = -5,4 \text{ kN}.$$

Hence,

$$R_B = R_A = 2,7 \text{ kN}.$$

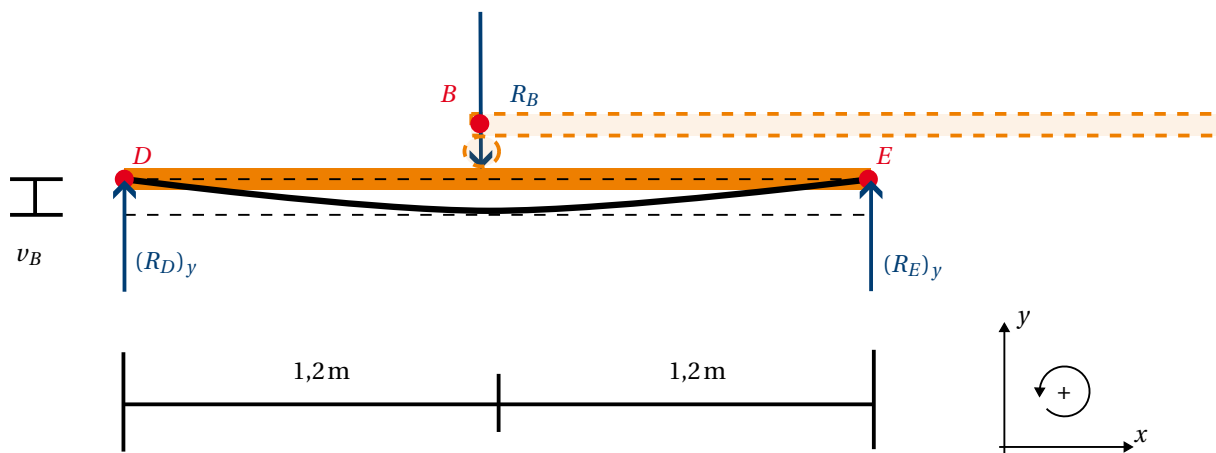


Figure 5.1: Deflection of the simply supported member DE due to the loading at point B

The deflection at the center point B of the simply supported beam DE shown on [Figure 5.1](#) is, per Appendix D:

$$v_B = -\frac{R_B L_{DE}^3}{48EI}.$$

Using the same line of reasoning as was the case in [Problem 16.46](#), the deflection at C will be exactly half of the deflection at B , as shown on [Figure 5.2](#), hence

$$v_{C,1} = \frac{1}{2} v_B = -\frac{R_B L_{DE}^3}{96EI}.$$

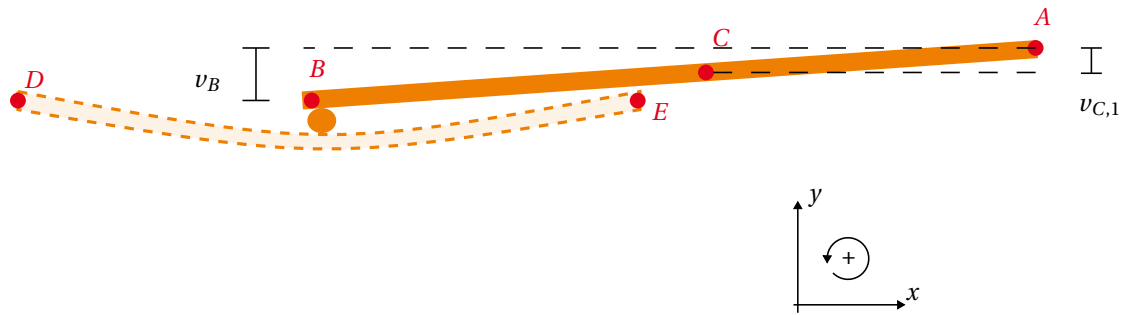
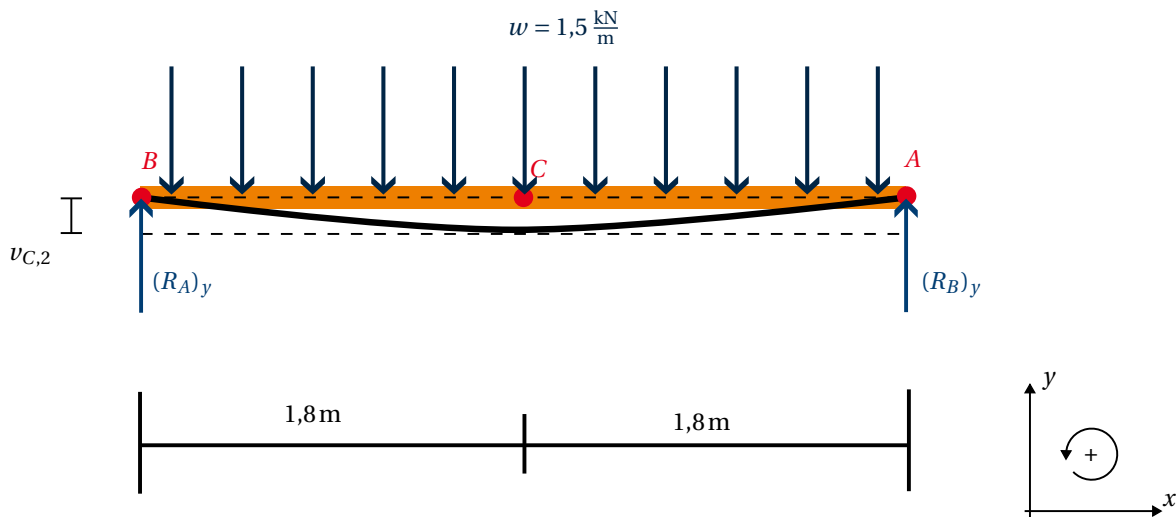


Figure 5.2: Rigid rotation of the member AB.

The deflection at the midpoint C (the maximum deflection) for the simply supported beam AB, depicted on Figure 5.3 is, per Appendix D:

$$v_{C,2} = -\frac{5wL_{AB}^4}{384EI}.$$

Figure 5.3: Deflection of the simply supported member AB due to the distributed loading w .

The moment of inertia around the neutral axis of each beam is

$$I = \frac{bh^3}{12} = \frac{75 \text{ mm} \cdot (150 \text{ mm})^3}{12} = 2,11 \cdot 10^{-5} \text{ m}^4.$$

Hence, the total displacement becomes:

$$\begin{aligned} v_{C,1} &= -\frac{R_B L_{DE}^3}{96EI} = -\frac{2,7 \text{ kN} \cdot (2,4 \text{ m})^3}{24 \cdot 10 \text{ GPa} \cdot 2,11 \cdot 10^{-5} \text{ m}^4} = -1,84 \cdot 10^{-3} \text{ m} \\ v_{C,2} &= -\frac{5wL_{AB}^4}{384EI} = -\frac{5 \cdot 1,5 \frac{\text{kN}}{\text{m}} \cdot (3,6 \text{ m})^4}{384 \cdot 10 \text{ GPa} \cdot 2,11 \cdot 10^{-5} \text{ m}^4} = -1,55 \cdot 10^{-2} \text{ m} \\ \Rightarrow v_C &= v_{C,1} + v_{C,2} = -1,84 \cdot 10^{-3} \text{ m} - 1,55 \cdot 10^{-2} \text{ m} = -17,3 \text{ mm}. \end{aligned}$$

$$v_C = -17,3 \text{ mm}$$

RESULT

Problem 12.33

The nails are on both sides of the cantilever beam and each can resist a shear of 2 kN. In addition to the distributed loading, determine the maximum load P that can be applied to the end of the beam. The nails are spaced 100 mm apart and the allowable shear stress for the wood is $\tau_{\text{allow}} = 3 \text{ MPa}$.

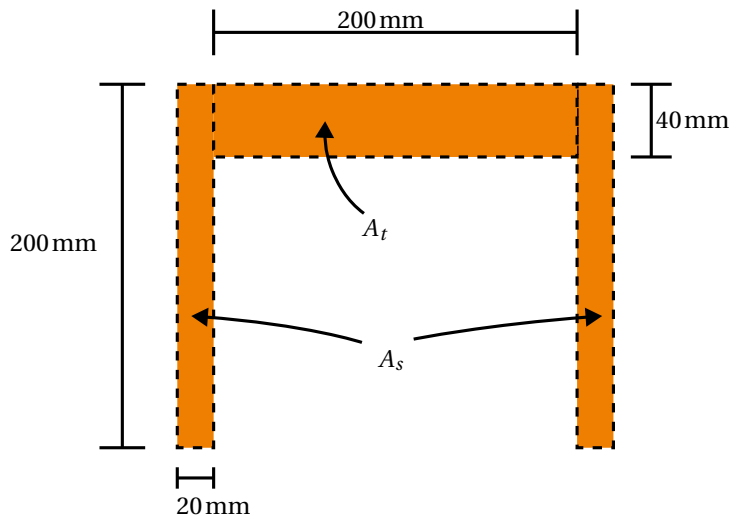
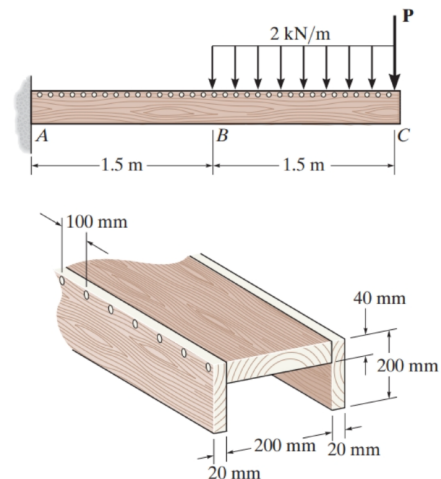


Figure 6.1: Cross section of the composite beam.

Derivation

From the figure, we see that the maximum shear occurs in the interval from A to B, where the shear is constant. Taking a force equilibrium in the y direction we obtain:

$$\begin{aligned}
 0 &= \sum F_y \\
 0 &= V_{\text{max}} - 2 \frac{\text{kN}}{\text{m}} \cdot 1,5 \text{ m} - P \\
 \Rightarrow V_{\text{max}} &= 3 \frac{\text{kN}}{\text{m}} + P.
 \end{aligned}$$

To find the centroid of the composite cantilever beam, we will consider it as consisting of three boards (two identical boards for the sides and a separate board for the top as depicted on [Figure 6.1](#)). The cross-sectional areas of these are, respectively:

$$\begin{aligned}
 A_s &= 20 \text{ mm} \cdot 200 \text{ mm} = 4 \cdot 10^3 \text{ mm}^2 \\
 A_t &= 40 \text{ mm} \cdot 200 \text{ mm} = 8 \cdot 10^3 \text{ mm}^2.
 \end{aligned}$$

Taking the bottom face of the beam as our origin, the formula for the centroid $\bar{y}$, can be used as:

$$\begin{aligned}\bar{y} &= \frac{\sum A_i y_i}{\sum A_i} \\ &= \frac{2 \cdot A_s \cdot y_s + A_t \cdot y_t}{A_s + A_t} \\ &= \frac{2 \cdot 4 \cdot 10^3 \text{ mm}^2 \cdot 100 \text{ mm} + 8 \cdot 10^3 \text{ mm}^2 \cdot 180 \text{ mm}}{2 \cdot 4 \cdot 10^3 \text{ mm}^2 + 8 \cdot 10^3 \text{ mm}^2} \\ &= 140 \text{ mm}.\end{aligned}$$

This is also shown on [Figure 6.2](#)

The moment of the area inducing shear is therefore:

$$Q = A_t \cdot (y_t - \bar{y}) = 8 \cdot 10^3 \text{ mm}^2 (180 \text{ mm} - 140 \text{ mm}) = 3,2 \cdot 10^5 \text{ mm}^3.$$

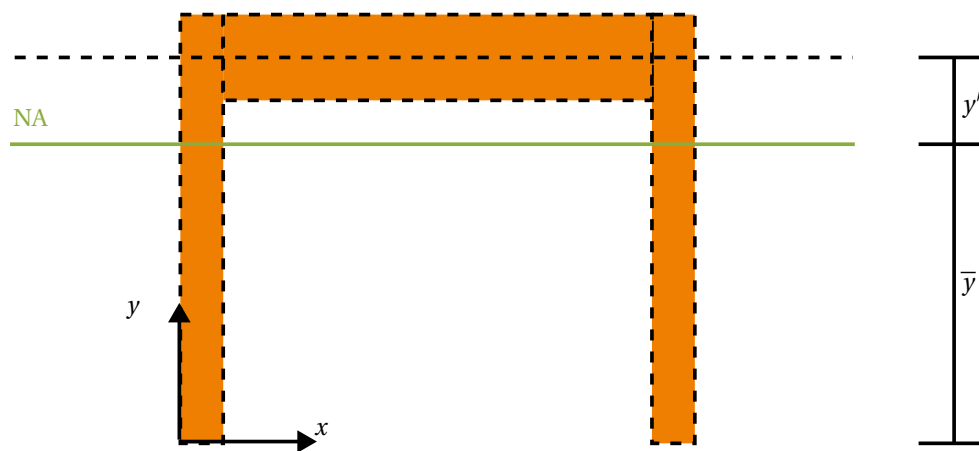


Figure 6.2: Cross section of the composite beam showing the position of the neutral axis, NA.

Now we must determine the moment of inertia I about the neutral axis. Using the general formula for the moment of inertia of a rectangle and the parallel-axis theorem, we can determine the moment of inertia about the neutral axis. This is done first for each side, as:

$$\begin{aligned}I_s &= \frac{1}{12} b_s h_s^3 + A_s d_s^2 \\ &= \frac{1}{12} \cdot 20 \text{ mm} \cdot (200 \text{ mm})^3 + 4 \cdot 10^3 \text{ mm}^2 \cdot (140 \text{ mm} - 100 \text{ mm})^2 \\ &= 1,97 \cdot 10^7 \text{ mm}^4.\end{aligned}$$

And now for the top plank as:

$$\begin{aligned}I_t &= \frac{1}{12} b_t h_t^3 + A_t d_t^2 \\ &= \frac{1}{12} \cdot 200 \text{ mm} \cdot (40 \text{ mm})^3 + 8 \cdot 10^3 \text{ mm}^2 (180 \text{ mm} - 140 \text{ mm})^2 \\ &= 1,39 \cdot 10^7 \text{ mm}^4.\end{aligned}$$

Summing the moments of inertia about the neutral axis of each of the constituent boards of the composite beam we obtain:

$$I = 2 \cdot I_s + I_t = 2 \cdot 1,97 \cdot 10^7 \text{ mm}^4 + 1,39 \cdot 10^7 \text{ mm}^4 = 5,33 \cdot 10^7 \text{ mm}^4.$$

Using the shear flow formula we find the maximum shear flow by using the previously found maximum shear

force.

$$q_{\max} = \frac{V_{\max} \cdot Q}{I}. \quad (6.1)$$

The nails can each hold $F = 2 \text{ kN}$ and are spaced by a distance of $s = 100 \text{ mm}$ between each pair. Hence, the allowable shear flow is:

$$q_{\text{allow}} = 2 \cdot \frac{2 \text{ kN}}{100 \text{ mm}} = 40 \frac{\text{kN}}{\text{m}}.$$

Inserting this into Equation (6.1) we can solve for the maximum allowable magnitude of the load P as:

$$\begin{aligned} q_{\text{allow}} &\geq q_{\max} \\ &\geq \frac{(3 \text{ kN} + P) \cdot Q}{I} \\ \Rightarrow P &\leq \frac{q_{\text{allow}} \cdot I}{Q} - 3 \text{ kN} \\ &\leq \frac{40 \frac{\text{kN}}{\text{m}} \cdot 5,33 \cdot 10^7 \text{ mm}^4}{3,2 \cdot 10^5 \text{ mm}^3} - 3 \text{ kN} \\ &\leq 3,67 \text{ kN}. \end{aligned}$$

This result implies that the nails are each capable of withstanding a maximum load of $P = 3,67 \text{ kN}$.

We must also check if this is within the allowable shear stress limit for the wood of $\tau_{\text{allow}} = 3 \text{ MPa}$. The largest vertical shear stress in the section will occur at the neutral axis. We have the formula for shear stress as:

$$\tau = \frac{VQ}{It}.$$

Here the shear $V = 3 \text{ kN} + 3,67 \text{ kN}$ is the shear at our limiting load of $P = 3,67 \text{ kN}$. Furthermore $t = 2 \cdot 20 \text{ mm}$ is the effective thickness where the shear is evaluated (twice the thickness of a single of the side boards). Lastly Q is the first moment of area at either side of the neutral axis (in this case we will use the bottom). The cross sectional areas of the boards below the neutral axis are:

$$A = 2 \cdot 20 \text{ mm} \cdot 140 \text{ mm}.$$

The distance from the centroid of this area to the neutral axis will be:

$$d = 140 \text{ mm} - 70 \text{ mm} = 70 \text{ mm}.$$

Hence;

$$Q = Ad = 2 \cdot 20 \text{ mm} \cdot 140 \text{ mm} \cdot 70 \text{ mm} = 3,92 \cdot 10^{-4} \text{ m}^3.$$

The shear stress is then:

$$\tau_{\max} = \frac{V \cdot Q}{I \cdot t} = \frac{(3 \text{ kN} + 3,67 \text{ kN}) \cdot 3,92 \cdot 10^{-4} \text{ m}^3}{5,33 \cdot 10^{-5} \text{ m}^4 \cdot 20 \text{ mm} \cdot 2} = 1,23 \text{ MPa}.$$

As $\tau_{\max} = 1,23 \text{ MPa} < \tau_{\text{allow}} = 3 \text{ MPa}$ the nails will be the limiting factor.

$P_{\max} = 3,67 \text{ kN}$

RESULT